

Understanding RWA, Impairment, and Risk-Sensitive Pricing

A Study Textbook on the A-IRB Risk-Weight Function,
IFRS 9 Expected Credit Losses, and Risk-Based Loan Pricing

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*Running example throughout this book:
an unsecured consumer instalment loan of €10,000.*

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Chapter 1

Introduction, Provenance, and How to Use This Book

1.1 Purpose and scope

A bank that grants a loan answers three quantitatively different questions about the same credit risk:

- (a) **Regulation (CRR/Basel):** *How much own funds must we hold against the unexpected loss of this loan?* The answer is expressed in risk-weighted assets (RWA) and derived from the A-IRB risk-weight function — the *Gordy formula*.
- (b) **Accounting (IFRS 9):** *How much expected loss must we recognise as an impairment allowance today?* The answer is the expected credit loss (ECL), possibly probability-weighted across macroeconomic scenarios.
- (c) **Management (pricing):** *What interest rate must we charge so that the loan earns its cost of capital after covering funding, operating cost, and expected loss?* The answer is the risk-sensitive break-even rate derived from a RAROC condition.

This book develops all three answers from first principles, at the level of a business-mathematics graduate (probability, statistics, financial mathematics): short proofs where feasible, proof sketches where a full proof would exceed the scope, worked numerical examples in every chapter, and references to the primary regulatory and academic sources. One single loan — an unsecured consumer instalment loan of €10,000 — serves as the running example throughout, so that the three frameworks can be compared on identical numbers.

Three companion Python scripts (`RWA_visualised.py`, `Impairment_visualised.py`, `Pricing_visualised.py`) implement the mathematics of Parts A–C as interactive Flask dashboards; Chapter 7 explains how to run them and how every number in this book can be reproduced with them.

1.2 How this book was created

This textbook was drafted by a large language model (AI) at the request of the reader, in an iterative process: the mathematical content follows published regulatory and academic sources (cited in the References and rated in the Source Notes); every numerical example was computed with executable code, not asserted; and two errors found in an earlier draft (an incorrect risk weight for the $PD = 8\%$ comparison and an inconsistent lifetime-EL pricing basis) were corrected against computation before publication. The prose was then revised for textbook-style exposition.

Disclaimers. The reader should treat this book accordingly:

1. This is an educational treatment with deliberate didactic simplifications. It is not legal, regulatory, accounting, tax, or investment advice.
2. The example parameters ($PD = 2\%$, $LGD = 45\%$, ...) are illustrative. A real A-IRB bank's parameters come from internally validated, supervisor-approved models and cannot be reconstructed from outside the bank.
3. The pricing model is a teaching model. Real banks add acquisition costs, liquidity costs, Pillar 2 add-ons, MREL and leverage-ratio constraints, competitive considerations, and behavioural effects.
4. Regulation changes. CRR amendments ("CRR3", output floors) and IFRS 9 interpretation practice evolve; always check the current consolidated text on EUR-Lex and the IFRS Foundation website.
5. Nothing in this book predicts whether any specific borrower receives a loan, or at what rate. The book explains how a bank *measures and prices* credit risk, not whether credit *is granted*.

1.3 Notation

Φ denotes the standard normal distribution function and $G = \Phi^{-1}$ its inverse (quantile function). $\mathbb{E}$ is expectation, $\mathbb{P}$ probability, Var variance. Currency amounts are euros. A full notation table is given in Appendix [A](#).

Chapter 2

Mathematical Foundations

This chapter collects the probabilistic and financial tools used later, so that the derivations in Parts A–C stay self-contained. Readers with a statistics background may skim it and return on demand.

2.1 The standard normal distribution and its quantiles

Definition 2.1 (Standard normal distribution). A random variable Z is *standard normally distributed* ($Z \sim \mathcal{N}(0, 1)$) if its density is

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R}, \quad (2.1)$$

with distribution function

$$\Phi(x) = \int_{-\infty}^x \varphi(z) dz. \quad (2.2)$$

Two numerical values recur throughout this book and should be memorised:

$$G(0.02) \approx -2.0537, \quad G(0.999) \approx 3.0902. \quad (2.3)$$

The first says: a 2% default probability corresponds to a default threshold 2.05 standard deviations *below* the mean of the credit-worthiness variable. The second says: the 99.9% quantile sits 3.09 standard deviations *above* the mean of the systematic factor — the supervisory stress state.

Lemma 2.2 (Linear combinations of independent normals). *Let Y, ε be independent with $Y, \varepsilon \sim \mathcal{N}(0, 1)$ and let $a, b \in \mathbb{R}$. Then*

$$X = aY + b\varepsilon \sim \mathcal{N}(0, a^2 + b^2). \quad (2.4)$$

Proof. The characteristic function of X is

$$\mathbb{E}[e^{it(aY+b\varepsilon)}] = \mathbb{E}[e^{itaY}] \mathbb{E}[e^{itb\varepsilon}] = e^{-(at)^2/2} e^{-(bt)^2/2} = e^{-(a^2+b^2)t^2/2}, \quad (2.5)$$

using independence and the characteristic function $e^{-t^2/2}$ of the standard normal. The result is the characteristic function of a $\mathcal{N}(0, a^2 + b^2)$ variable, and characteristic functions determine the distribution uniquely. $\square$

2.2 Conditional probability and conditional expectation

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (2.6)$$

If $X = aY + b\varepsilon$ and we *condition on* the realised value $Y = y$, then X becomes a deterministic shift plus noise:

$$X | Y = y \sim \mathcal{N}(ay, b^2). \quad (2.7)$$

Equation (2.7) is the engine of the entire A-IRB derivation: conditioning on a bad economy shifts the mean of every borrower's credit-worthiness downward and thereby raises every borrower's default probability.

2.3 Expectation, variance decomposition, and tail risk

For a loss variable $L \geq 0$,

$$\text{EL} = \mathbb{E}[L] \quad (\text{expected loss}), \quad (2.8)$$

while economic or regulatory capital addresses the *upper tail*:

$$\text{capital} \approx \text{tail quantile of } L - \text{EL}. \quad (2.9)$$

This “tail minus expectation” logic is the single most important structural idea of this book:

Key insight. Expected loss is a *cost of doing business* — it is priced into the interest rate and provisioned under IFRS 9. *Unexpected* loss — the deviation beyond expectation in a severe scenario — is what regulatory capital must absorb. Part A computes the tail term, Part B the expectation term, and Part C charges both to the customer.

2.4 Present value and the annuity

A cash flow C received at time t has present value $C(1+r)^{-t}$ at discount rate r . For a loan of principal N , nominal rate r , and term n years with constant annual annuity A , the condition $\text{PV} = N$ gives

$$A = \frac{Nr}{1 - (1+r)^{-n}}. \quad (2.10)$$

Example 2.3 (Running example, annuity). With $N = €10,000$, $r = 8.5\%$, $n = 3$:

$$A = \frac{10,000 \times 0.085}{1 - 1.085^{-3}} = €3,915.39. \quad (2.11)$$

The amortisation schedule (used in Parts B and C) is shown in Table 2.1.

Table 2.1: Amortisation schedule of the running example (€10,000, 8.5%, 3 years, annuity €3,915.39).

Year	Balance (BoY)	Interest	Principal	Balance (EoY)
1	10,000.00	850.00	3,065.39	6,934.61
2	6,934.61	589.44	3,325.95	3,608.66
3	3,608.66	306.74	3,608.66	0.00

Chapter 3

Part A — Risk-Weighted Assets under the A-IRB Approach

3.1 The regulatory question

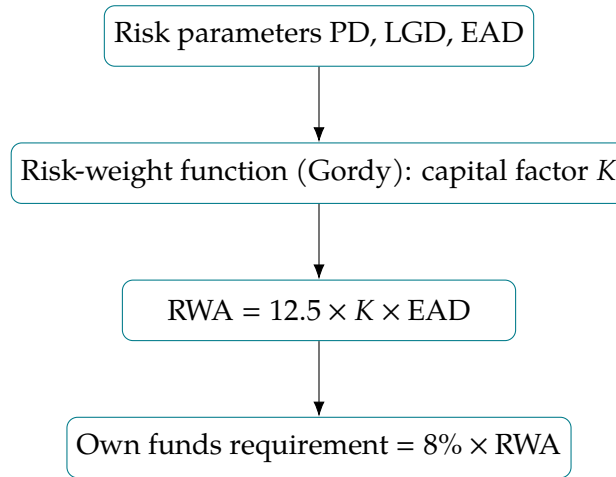


Figure 3.1: The A-IRB computation chain for a single exposure.

Pillar 1 of the capital framework requires banks to hold own funds of at least 8% of their risk-weighted assets. Two building blocks therefore have to be understood: *risk-weighted assets* and the 8% rule.

Definition 3.1 (Risk-weighted assets). The risk-weighted assets of an exposure are its exposure value EAD multiplied by a *risk weight* RW:

$$\text{RWA} = \text{RW} \times \text{EAD}. \quad (3.1)$$

In the standardised approach RW is a fixed supervisory percentage; in the IRB approaches RW is *computed* from risk parameters.

3.2 The factor 12.5: a complete derivation

Theorem 3.2 (RWA scaling). Let the minimum own funds ratio be 8%, and let the capital requirement for one exposure be $K \times \text{EAD}$ (with K derived from the risk-weight function). Then the risk-weighted assets that make the two statements consistent are

$$\text{RWA} = \frac{K \times \text{EAD}}{0.08} = 12.5 \times K \times \text{EAD}. \quad (3.2)$$

Proof. By definition of the capital ratio, the requirement is

$$\text{own funds} = 0.08 \times \text{RWA}. \quad (3.3)$$

The IRB capital requirement for the exposure is $\text{own funds} = K \times \text{EAD}$. Equating and dividing by 0.08 ($1/0.08 = 12.5$) yields (3.2). $\square$

This is pure algebra — there is no economics in the 12.5; the economics sits entirely inside K , to which we now turn.

3.3 The three risk parameters

PD (probability of default). The probability that the borrower defaults within one year under the supervisory default definition (CRR Art. 178: 90 days past due or unlikeliness to pay). Running example: PD = 2%.

LGD (loss given default). The expected loss as a fraction of the exposure *given* default, net of recoveries, including workout costs and discounting of recoveries (CRR Art. 181). Unsecured consumer loans have no collateral, but collections recoveries still reduce the loss. Running example: LGD = 45%.

EAD (exposure at default). The expected outstanding amount at the moment of default. For a fully drawn instalment loan the EAD approximates the outstanding balance; immediately after disbursement, $\text{EAD} \approx e10,000$. Under A-IRB the bank estimates EAD internally.

Remark 3.3 (F-IRB versus A-IRB). Under the Foundation IRB approach the bank estimates only PD; LGD and EAD follow supervisory values. Under the *Advanced* IRB approach the bank estimates PD, LGD, and EAD with internal, supervisor-approved models. The risk-weight function itself is prescribed — the “A” refers to the parameter estimation, not to the formula.

3.4 The single-factor model: from Merton to Vasicek

The IRB formula is not an arbitrary regulatory curve. Its intellectual pedigree is Merton (1974) → Vasicek (2002) → Gordy (2003) → Basel II/III IRB → CRR Arts. 153–154. We derive it in four steps.

3.4.1 Step 1: the factor model

Definition 3.4 (Asymptotic single risk factor model). For each borrower i , the standardised asset (credit-worthiness) return is

$$X_i = \sqrt{R} Y + \sqrt{1 - R} \varepsilon_i, \quad (3.4)$$

where $Y \sim \mathcal{N}(0, 1)$ is a single systematic factor (the economy), $\varepsilon_i \sim \mathcal{N}(0, 1)$ is idiosyncratic noise, all variables mutually independent, and $R \in (0, 1)$ is the asset correlation.

Lemma 3.5 (X_i is standard normal). $X_i \sim \mathcal{N}(0, 1)$.

Proof. By Lemma 2.2 with $a = \sqrt{R}$, $b = \sqrt{1 - R}$: $\text{Var}(X_i) = R + (1 - R) = 1$ and mean 0; normality follows from the linear combination of independent Gaussians. $\square$

Interpretation: a borrower’s fortunes are driven by the macroeconomy (weight $\sqrt{R}$) and by personal circumstances (weight $\sqrt{1 - R}$). The correlation of any two borrowers’ returns is

$$\text{Cov}(X_i, X_j) = R \text{Var}(Y) = R, \quad (3.5)$$

which explains the name “asset correlation” for R .

3.4.2 Step 2: the default threshold

Definition 3.6 (Merton default rule). Borrower i defaults in the period iff $X_i < c$, where the threshold c is chosen such that the *unconditional* default probability equals the PD:

$$\mathbb{P}(X_i < c) = \text{PD}. \quad (3.6)$$

Since $X_i \sim \mathcal{N}(0, 1)$, the threshold is

$$\Phi(c) = \text{PD} \iff c = G(\text{PD}) = \Phi^{-1}(\text{PD}). \quad (3.7)$$

For $\text{PD} = 2\%$: $c = G(0.02) = -2.0537$. Default occurs 2.05 standard deviations below average fortune — a rare event in good times.

3.4.3 Step 3: the conditional default probability

Now condition on the economy being in a specific state $Y = y$. Using (2.7) with $a = \sqrt{R}$, $b = \sqrt{1-R}$:

$$X_i | Y = y \sim \mathcal{N}(\sqrt{R}y, 1-R), \quad (3.8)$$

hence

$$\mathbb{P}(\text{default} | Y = y) = \mathbb{P}(\sqrt{R}y + \sqrt{1-R}\varepsilon < G(\text{PD})) \quad (3.9)$$

$$= \mathbb{P}\left(\varepsilon < \frac{G(\text{PD}) - \sqrt{R}y}{\sqrt{1-R}}\right) = \Phi\left(\frac{G(\text{PD}) - \sqrt{R}y}{\sqrt{1-R}}\right). \quad (3.10)$$

The regulator fixes the adverse systematic state at the 99.9% quantile. With the CRR sign convention (adverse state in the upper tail),

$$y^{\text{stress}} = G(0.999) \approx 3.0902, \quad (3.11)$$

and the *stress default probability* becomes

$$\boxed{\text{PD}^{\text{stress}} = \Phi\left[\frac{1}{\sqrt{1-R}} G(\text{PD}) + \sqrt{\frac{R}{1-R}} G(0.999)\right]} \quad (3.12)$$

— the term in square brackets is simply (3.10) with $y = G(0.999)$, split over the common denominator $\sqrt{1-R}$.

Remark 3.7 (Why the stress PD can be many times the PD). The correlation term multiplies $G(0.999) \approx 3.09$ by $\sqrt{R/(1-R)}$. For $R \approx 9.5\%$ this contributes roughly +1.0 to the argument of Φ . Near the far left tail of Φ , one unit of argument is worth a *multiplication* of the tail probability: with $\text{PD} = 2\%$ the stress PD is 12.3% — six times higher. This systematic amplification is precisely the economic content of the correlation.

3.4.4 Step 4: capital equals stress loss minus expected loss

The loss rate of the exposure, *given* default, is LGD. In the stress state the expected loss rate is therefore $\text{LGD} \times \text{PD}^{\text{stress}}$; the unconditional expected loss rate is $\text{EL} = \text{PD} \times \text{LGD}$. Capital covers the *unexpected* part:

$$\boxed{K = \text{LGD} \times \Phi\left[\frac{1}{\sqrt{1-R}} G(\text{PD}) + \sqrt{\frac{R}{1-R}} G(0.999)\right] - \text{PD} \times \text{LGD}} \quad (3.13)$$

This is the **Gordy formula**. Factoring out LGD gives the transparent form $K = \text{LGD} \times (\text{PD}^{\text{stress}} - \text{PD})$: capital equals the loss-given-default times the *excess* of the stress default probability over the ordinary one.

3.5 Why the formula applies loan-by-loan: Gordy's theorem

Regulation applies (3.13) exposure by exposure. That is only legitimate because of a remarkable asymptotic property.

Theorem 3.8 (Gordy (2003), portfolio invariance — proof sketch). *Consider a portfolio of n exposures of equal size, each following (3.4) with the same R and independent ε_i . Conditionally on Y , defaults are independent, each with probability $PD^{stress}(Y)$. By the law of large numbers, as $n \rightarrow \infty$ the portfolio default rate converges to $PD^{stress}(Y)$ almost surely; idiosyncratic risk vanishes. The portfolio loss rate at the 99.9% quantile of Y is therefore exactly $LGD \times PD^{stress}$, with no residual idiosyncratic term, and the capital per unit of exposure depends only on the exposure's own (PD, LGD) and on the common R — not on the portfolio's composition.*

The full proof (Gordy 2003) requires the uniform-exposure-size and single-factor assumptions; both are idealisations. Real portfolios are finite and concentrated, which is why banks hold granularity buffers and why economic capital (Chapter 5) exceeds regulatory capital. But the theorem explains the structural feature that makes the IRB usable: *capital is additive because idiosyncratic risk is diversified away in the limit, and what remains is systematic risk, priced identically for every exposure with the same parameters.*

3.6 The CRR retail correlation function

For retail exposures, the CRR (Art. 154) does not let the bank choose R freely. It prescribes a decreasing function of the PD. Writing

$$A(PD) = \frac{1 - e^{-35PD}}{1 - e^{-35}}, \quad (3.14)$$

the correlation is

$$R(PD) = 0.03 \times A(PD) + 0.16 \times (1 - A(PD)) = 0.16 - 0.13 \times A(PD) \quad (3.15)$$

Interpretation of the two anchors:

- $PD \rightarrow 0$: $A \rightarrow 0$, hence $R \rightarrow 16\%$. Very safe borrowers are driven almost entirely by long macro cycles — think of prime mortgage borrowers who only default in system-wide crises.
- $PD \rightarrow 1$: $A \rightarrow 1$, hence $R \rightarrow 3\%$. Already-distressed borrowers default for idiosyncratic reasons; the economy matters less for *whether* they default.

For special retail sub-classes the CRR replaces (3.15) by constants: $R = 0.15$ for exposures secured by immovable property, and $R = 0.04$ for qualifying revolving retail exposures. Our unsecured instalment loan uses the general function (3.15).

3.7 Worked example: the €10,000 loan, step by step

Assumptions (illustrative): $EAD = €10,000$, $PD = 2\%$, $LGD = 45\%$, non-revolving unsecured retail.

Step 1 — Correlation. With $35 \times PD = 0.7$:

$$A = \frac{1 - e^{-0.7}}{1 - e^{-35}} = 0.5034, \quad (3.16)$$

$$R = 0.16 - 0.13 \times 0.5034 = 0.0946 \quad (9.46\%). \quad (3.17)$$

Step 2 — Quantiles.

$$G(0.02) = -2.0537, \quad G(0.999) = 3.0902. \quad (3.18)$$

Step 3 — Stress argument.

$$z = \frac{-2.0537}{\sqrt{1 - 0.0946}} + \sqrt{\frac{0.0946}{1 - 0.0946}} \times 3.0902 = -2.1561 + 1.0000 = -1.1597. \quad (3.19)$$

Step 4 — Stress PD.

$$PD^{\text{stress}} = \Phi(-1.1597) = 12.31\%. \quad (3.20)$$

A once-in-a-thousand-years downturn raises this borrower's default probability from 2% to 12.3%.

Step 5 — Capital factor.

$$K = 0.45 \times (0.1231 - 0.02) = 0.45 \times 0.1031 = 0.0464 \quad (4.64\%). \quad (3.21)$$

Step 6 — RWA and own funds.

$$RW = 12.5 \times K = 58.0\%, \quad (3.22)$$

$$RWA = 0.5799 \times 10,000 = e5,798.64, \quad (3.23)$$

$$\text{own funds} = 0.08 \times 5,798.64 = e463.89. \quad (3.24)$$

Table 3.1: Part A summary for the running example.

Quantity	Value
PD / LGD / EAD	2% / 45% / €10,000
Correlation R	9.46%
Stress PD (99.9%)	12.31%
Capital factor K	4.64%
Risk weight RW	58.0%
RWA	€5,798.64
Expected loss $PD \times LGD \times EAD$	€90.00
Own funds requirement ($8\% \times RWA$)	€463.89

Key insight. The bank neither expects to lose €5,798.64 nor €463.89. It *expects* €90 of loss (covered by pricing and provisions) and must *hold* €463.89 of capital against the 99.9% tail deviation beyond that expectation.

3.8 Sensitivities

Table 3.2 varies the PD; Table 3.3 the LGD. The required rate column anticipates Chapter 5 (12-month EL basis, funding 2%, opex €166.67 p.a., hurdle 12%, regulatory capital).

Remark 3.9 (Reading the PD table). K rises with PD but *less than proportionally*, because the correlation falls as PD rises (16% $\rightarrow$ 3%), cushioning the stress amplification. The risk weight is thus concave in PD. RWA is linear in EAD and LGD. The pricing-relevant quantity, however, is dominated by expected loss: between PD = 2% and PD = 8% the own funds requirement grows by only 23% while the EL grows by a factor of four — a first glimpse of why risk-based pricing is driven by Part B more than by Part A.

Table 3.2: PD sensitivity (LGD = 45%, EAD = €10,000).

PD	R	PD ^{stress}	K	RW	RWA	own funds	req. rate
0.5%	13.9%	6.25%	2.59%	32.4%	3,236	258.89	5.22%
1.0%	12.2%	9.14%	3.66%	45.8%	4,577	366.18	5.73%
2.0%	9.5%	12.31%	4.64%	58.0%	5,799	463.89	6.56%
5.0%	5.3%	16.81%	5.31%	66.4%	6,642	531.32	8.65%
8.0%	3.8%	20.64%	5.69%	71.1%	7,108	568.64	10.69%
15%	3.1%	30.75%	7.09%	88.6%	8,860	708.81	15.53%
30%	3.0%	50.44%	9.20%	115.0%	11,498	919.82	25.76%

Table 3.3: LGD sensitivity (PD = 2%, EAD = €10,000).

LGD	K	RWA	EL (p.a.)
10%	1.03%	1,288.59	20.00
25%	2.58%	3,221.47	50.00
45%	4.64%	5,798.64	90.00
65%	6.70%	8,375.82	130.00
90%	9.28%	11,597.29	180.00

3.9 Limitations of Part A

- Real A-IRB banks face PD floors, downturn-LGD requirements, CCF/EAD model risk, and (since CRR2) an output floor tying IRB RWA to a percentage of standardised RWA — the applicable percentage depends on the bank's transition schedule and current CRR state.
- The ASRF assumptions (infinite granularity, single factor) fail for concentrated portfolios; supervisory add-ons and Pillar 2 address this.
- The 99.9% level is a calibration choice, not an empirical fact about economies.

Chapter 4

Part B — Impairment under IFRS 9

4.1 The accounting question

Accounting asks a different question than regulation. Regulation asks how much capital to hold against unexpected loss; IFRS 9 asks how much *expected* loss to recognise *today* as an allowance (impairment) in profit or loss. IFRS 9 (effective 2018) replaced the IAS 39 incurred-loss model — under which losses were recognised only when a triggering event had occurred — with a forward-looking expected-credit-loss model. The philosophical change: loss recognition moved from *evidence of damage* to *probability-weighted forecast*.

4.2 The three-stage model

IFRS 9 (paras 5.5.15–5.5.20) sorts financial assets into three stages. The classification governs two things: the *measurement horizon* of the allowance and the *interest recognition basis*.

Stage	Trigger	Allowance / interest basis
1	no significant deterioration	12-month ECL; interest gross
2	significant increase in credit risk	lifetime ECL; interest gross
3	credit-impaired (default-like)	lifetime ECL; interest net

The 12-month/lifetime distinction is the *cliff* of the model. Crossing a stage-2 trigger multiplies the allowance for a multi-period loan, which is why stage migration — not default itself — is a major driver of bank P&L volatility in downturns.

Definition 4.1 (Significant increase in credit risk (SICR)). A stage-2 migration occurs when the lifetime PD at reporting rises significantly above the lifetime PD expected at initial recognition. Common operationalisations: 30+ days past due (rebuttable presumption), internal rating deterioration by a defined number of notches, or lifetime PD more than doubling versus origination.

4.3 The ECL formula

For a loan with amortised-cost balance E_t at the start of period t , marginal one-year default probability m_t (conditional on survival to t), and loss given default LGD:

$$\text{ECL} = \sum_{t=1}^T m_t \times \text{LGD} \times E_t \times (1 + \text{EIR})^{-t} \quad (4.1)$$

Three properties are prescribed by IFRS 9.5.5.17 and deserve emphasis:

1. **Probability-weighted and forward-looking.** The ECL must be a probability-weighted average over multiple macroeconomic scenarios — not a single “most likely” path, and not a worst case.
2. **Discounted at the original EIR.** The allowance is a present value; a loss expected in year 3 is discounted at the loan’s own effective interest rate.
3. **Marginal, not cumulative, PDs.** The t -th term uses the probability of defaulting *in* period t given survival until then. With survival factors $s_t = \prod_{j \leq t} (1 - m_j)$, the year- t term can equivalently be written with cumulative PDs, $(S_t - S_{t-1})$, where $S_t = 1 - s_t$.

Remark 4.2. The triad $m_t \times \text{LGD} \times E_t$ is the same credit-risk arithmetic as the regulatory $\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}$. The frameworks differ in *timing, horizon, aggregation, and discounting* — not in the loss building blocks. This is deliberate in this book: the same parameters feed both models, which is exactly what the companion dashboards demonstrate.

4.4 A short proof: lifetime dominates 12-month ECL

Theorem 4.3 (Monotonicity of the allowance in the horizon). *For a loan with remaining life $T \geq 2$ and non-negative marginal PDs,*

$$\text{lifetime ECL} \geq \text{12-month ECL}, \quad (4.2)$$

with equality iff $m_t = 0$ for all $t \geq 2$ (or $T \leq 1$).

Proof. The 12-month ECL is the $t = 1$ term of the sum (4.1); the lifetime ECL is the sum of the terms $t = 1, \dots, T$. Every term is a product of non-negative factors, so adding terms $t \geq 2$ cannot decrease the total; it strictly increases it if any $m_t > 0$ for $t \geq 2$. $\square$

Corollary 4.4. *Stage migration from stage 1 to stage 2 never decreases the allowance of a multi-period performing loan. This is the analytical origin of the P&L cliff effect.*

4.5 Worked example: the €10,000 loan

Loan terms: €10,000, 3-year annuity, EIR 8.5% (schedule in Table 2.1). Marginal PDs (base scenario, conditional on survival): $m = (2.0\%, 2.5\%, 3.0\%)$; $\text{LGD} = 45\%$.

4.5.1 Stage 1: 12-month ECL

$$\text{loss year 1 (undiscounted)} = 0.02 \times 0.45 \times 10,000 = e90.00, \quad (4.3)$$

$$\text{ECL}_{12\text{m}} = \frac{90.00}{1.085} = e82.95. \quad (4.4)$$

4.5.2 Lifetime ECL, base scenario

4.5.3 Scenario weighting

IFRS 9.5.5.17 requires probability weighting across scenarios. We use three: base (60%), optimistic with halved PDs (20%), adverse with doubled PDs (20%). Because every term of (4.1) is linear in the m_t , each scenario’s ECL scales linearly in its PD scale:

$$\text{ECL}_{\text{opt}} = 93.68, \quad \text{ECL}_{\text{base}} = 187.36, \quad \text{ECL}_{\text{adv}} = 374.72, \quad (4.5)$$

$$\text{ECL}_{\text{lt}} = 0.2 \times 93.68 + 0.6 \times 187.36 + 0.2 \times 374.72 = e206.10. \quad (4.6)$$

Table 4.1: Lifetime ECL decomposition, base scenario.

Year t	m_t	E_t	$m_t \text{LGDE}_t$	PV at 8.5%
1	2.0%	10,000.00	90.00	82.95
2	2.5%	6,934.61	78.01	66.27
3	3.0%	3,608.66	48.72	38.14
Lifetime ECL (base)				187.36

Table 4.2: Scenario-weighted lifetime ECL under alternative weightings.

Weighting (opt/base/adv)	Optimistic share	Adverse share	ECL
20/60/20 (baseline)	20%	20%	206.10
33/34/33	33%	33%	218.27
10/80/10	10%	10%	196.73
50/0/50	50%	50%	234.20

The same weighting applied to the stage-1 horizon gives the probability-weighted 12-month ECL $0.2 \times 41.47 + 0.6 \times 82.95 + 0.2 \times 165.90 = €91.24$ (the companion dashboard displays these weighted figures).

Key insight. The weighting is linear in the scenario PDs, so the *choice of weights* matters less than the *shape of the scenario fan*. The non-linearity of real ECL models comes from PD responses to macro drivers inside each scenario, not from the averaging step.

4.5.4 Stage-2 migration: the cliff in numbers

Suppose that after one year the borrower's credit risk has deteriorated such that the remaining lifetime PD has roughly doubled (a plausible SICR trigger). The loan migrates from stage 1 to stage 2. For the remaining periods the allowance becomes the lifetime figure on the then-outstanding balance of €6,934.61 — about €104 for years 2–3 ($66.27 + 38.14$, base scenario) versus €82.95 under a fresh 12-month view. Nothing about the contractual cash flows has changed; only the *expected timing* of losses. A bank with a large book migrating simultaneously books a large impairment charge — the mechanism behind the P&L shocks of 2020.

4.5.5 The discounting subtlety

Because the ECL is discounted at the loan's own EIR, a *higher* contract rate lowers the discounted allowance:

EIR	5%	8.5%	12%	20%
12-month ECL	85.71	82.95	80.36	75.00

All else equal, a riskier-priced loan (higher EIR) shows a slightly *smaller* allowance for the same expected loss — a real, disclosed, and sometimes counter-intuitive effect of the standard.

4.6 Reconciliation with Part A

- Both frameworks use PD, LGD, and an exposure notion. The regulatory year-1 expected loss (€90) equals the undiscounted first-period IFRS 9 loss in the base scenario — deliberately, because the same parameters feed both models.

- Regulation holds capital against the *tail* ($K \times \text{EAD} = e463.89$ covering the 99.9% conditional loss *beyond* expectation); IFRS 9 books the *expectation* (€82.95–€206.10) as a provision. Both coexist: an A-IRB bank books IFRS 9 ECL as a provision and additionally holds IRB capital.
- Under CRR Arts. 158 and 160–162, the excess of IRB expected losses over accounting provisions (or vice versa) enters the own funds calculation. The details are bank-specific and outside our scope.

4.7 Limitations of Part B

- Real banks run score-to-PD models with macroeconomic overlays and dozens of scenarios; three scenarios are the minimum acceptable teaching device.
- SICR assessment is partly judgemental; the “lifetime PD doubling” rule is market practice, not a standard requirement.
- IFRS 9 exposure is the amortised-cost carrying amount (plus, for revolvers, expected drawdowns), which need not equal the regulatory EAD.
- Stage 3 (credit-impaired) measurement — the probability-weighted shortfall of cash flows — is not modelled here.
- A further subtlety: marginal PDs conditioned on survival make the ECL depend on the *whole* PD term structure, not just on the first-year PD.

Chapter 5

Part C — Risk-Sensitive Pricing

5.1 The pricing question

Given the risk of the loan (Part A) and its expected loss and provisioning (Part B), what interest rate should the bank charge so that the loan earns its cost of capital? This is risk-based pricing: prices that reflect the borrower's risk profile — standard practice in consumer and mortgage lending.

5.2 The RAROC framework

Risk-Adjusted Return on Capital (RAROC), developed at Bankers Trust in the late 1970s, allocates the loan's risk capital — here the regulatory own funds requirement — and compares the loan's risk-adjusted profit with the shareholders' hurdle rate (cost of equity, COE):

$$\text{RAROC} = \frac{\text{revenue} - \text{cost of funds} - \text{operating cost} - \text{expected loss}}{\text{capital}}. \quad (5.1)$$

Definition 5.1 (Pricing rule). Accept or price the loan such that $\text{RAROC} \geq \text{COE}$. The *break-even rate* is the customer rate r_L that makes $\text{RAROC} = \text{COE}$ exactly.

5.3 Derivation of the required rate

Let B be the average outstanding balance over the life of the loan, r_L the customer rate, r_F the funding rate, O the amortised operating cost per year, $\text{EL}_{\text{p.a.}}$ the expected loss charged to pricing per year, $C = 0.08 \times \text{RWA}$ the allocated capital, and h the hurdle rate. Setting (5.1) equal to h :

$$(r_L - r_F) B - O - \text{EL}_{\text{p.a.}} = h C \quad (5.2)$$

$$\boxed{r_L = r_F + \frac{O + \text{EL}_{\text{p.a.}} + h C}{B}} \quad (5.3)$$

Equation (5.3) is the *additive pricing formula*: customer rate = funding cost + operating cost + expected loss + cost of capital, each per unit of exposure. The third term converts Part B into price; the fourth converts Part A into price.

5.4 Worked example: pricing the €10,000 loan

Assumptions (illustrative, per year unless stated):

- funding rate $r_F = 2.0\%$;

- one-time origination/servicing cost €500, amortised over 3 years: $O = e166.67$ p.a.;
- expected loss on the 12-month basis: $EL_{p.a.} = PDLGD EAD = e90.00$;
- capital $C = e463.89$ (Part A); hurdle rate $h = 12\%$;
- average outstanding balance over 3 years (from Table 2.1): $B = (10,000 + 6,934.61 + 3,608.66)/3 = e6,847.75$.

$$r_L = 0.02 + \frac{166.67 + 90.00 + 0.12 \times 463.89}{6,847.75} = 0.02 + \frac{312.34}{6,847.75} = 6.56\% \text{ p.a.} \quad (5.4)$$

Table 5.1: Decomposition of the 4.56% risk-and-cost margin.

Component	Margin share	Basis
Operating cost	2.43%	$166.67/6,847.75$
Expected loss	1.31%	$90.00/6,847.75$
Cost of capital	0.81%	$0.12 \times 463.89/6,847.75$
Total margin	4.56%	

5.5 Cross-check: RAROC at the example EIR of 8.5%

If the bank charges 8.5% (the rate assumed for the amortisation schedule in Part B), the RAROC is

$$\text{RAROC}(8.5\%) = \frac{(0.085 - 0.02) \times 6,847.75 - 166.67 - 90.00}{463.89} = \frac{188.44}{463.89} = 40.6\%. \quad (5.5)$$

A 40% RAROC looks implausibly high — and this is the model’s most important lesson: *regulatory capital for a low-PD retail loan is small*, so a RAROC computed on regulatory capital is easily inflated. Professional pricing desks therefore use economic capital (a bank-specific tail measure, often several times the regulatory figure for undiversified retail books), Pillar 2 add-ons, MREL and leverage-ratio constraints, and acquisition costs that dwarf servicing costs. The break-even logic (5.3) is unchanged; only the size of C and O grows. Table 5.2 shows the effect of the two most important corrections.

Table 5.2: Break-even rate (%) as a function of hurdle rate and capital multiplier (PD = 2%, 12-month and lifetime EL basis).

Capital multiplier	12-month EL basis			lifetime EL basis		
	$h = 8\%$	$h = 12\%$	$h = 16\%$	$h = 8\%$	$h = 12\%$	$h = 16\%$
1.0 (regulatory)	6.29	6.56	6.83	5.98	6.25	6.52
2.0	6.83	7.37	7.92	6.52	7.06	7.60
4.0 (econ. approx.)	7.92	9.00	10.08	7.60	8.69	9.77

Key insight. With economic capital of four times regulatory capital and a 12% cost of equity, the teaching model’s break-even rate (9.00%) lands in the region of observed market rates for unsecured consumer loans — the model closes the gap between regulatory arithmetic and observed prices. Use the multiplier slider in `Pricing_visualised.py` to explore this.

5.6 Risk sensitivity: two borrowers, same product

Take a riskier borrower with $PD = 8\%$, $LGD = 45\%$ (row of Table 3.2): $R = 3.79\%$, $PD^{\text{stress}} = 20.64\%$, $K = 5.69\%$, $RW = 71.1\%$, $RWA = \text{€}7,108$, own funds $\text{€}568.64$, and an EL of $0.08 \times 0.45 = \text{€}360$ p.a. The required rate becomes

$$r_L = 0.02 + \frac{166.67 + 360.00 + 0.12 \times 568.64}{6,847.75} = 10.69\% \text{ p.a.}, \quad (5.6)$$

against 6.56% for the 2% borrower. Note the composition: regulatory capital rises only moderately (the falling correlation cushions the stress PD), while the *expected loss* term does most of the pricing work. Risk-based pricing in retail is, in this precise sense, mostly expected-loss pricing plus a modest capital charge.

5.7 Pricing with a lifetime-EL view

Part B showed the probability-weighted lifetime ECL ($\text{€}206.10$) is a one-off present value, while the recurring 12-month EL charged every year ($\text{€}90$ p.a. $\times 3$ years, ignoring run-off) overstates total expected losses. If management prices off the amortised lifetime view, the EL term becomes $206.10/3 = \text{€}68.70$ p.a., lowering the required rate to about 6.25% (Table 5.2, right panel). The choice of EL basis — recurring 12-month versus amortised lifetime — is a genuine, disclosed pricing-policy decision, not a technicality: the 12-month basis is the conservative one because it charges a full-EAD expected loss even as the balance amortises.

5.8 Limitations of Part C

- RAROC on regulatory capital is biased upward for low-risk retail; economic capital, RAROC, or EVA are the professional fixes.
- The model ignores taxes, the liquidity cost of the funding base, prepayment risk (consumer loans prepay), cross-selling subsidies, adverse-selection feedback (raising the price worsens the retained pool), and regulatory price constraints.
- A single hurdle rate ignores portfolio context: a new loan is priced on *marginal* capital, not average capital.

Chapter 6

Synthesis: One Loan, Three Numbers

Table 6.1: The running example across the three frameworks (base case).

Question	Framework	Number
Unexpected tail-loss capital	CRR A-IRB (Part A)	€463.89 capital (RWA = €5,798.64)
Expected-loss provision	IFRS 9 (Part B)	€82.95 (stage 1) to €206.10 (stage 2, weighted life)
Break-even customer rate	RAROC pricing (Part C)	6.56% p.a. (12-month EL basis)

The three numbers are mechanically linked: the same PD, LGD, and exposure drive all three. Change the PD from 2% to 8% and all three move — the capital requirement rises by about a quarter (correlation cushioning), the stage-1 allowance roughly quadruples, and the required rate rises by about two thirds, driven mainly by expected loss. This coupling is exactly what the three companion dashboards visualise interactively.

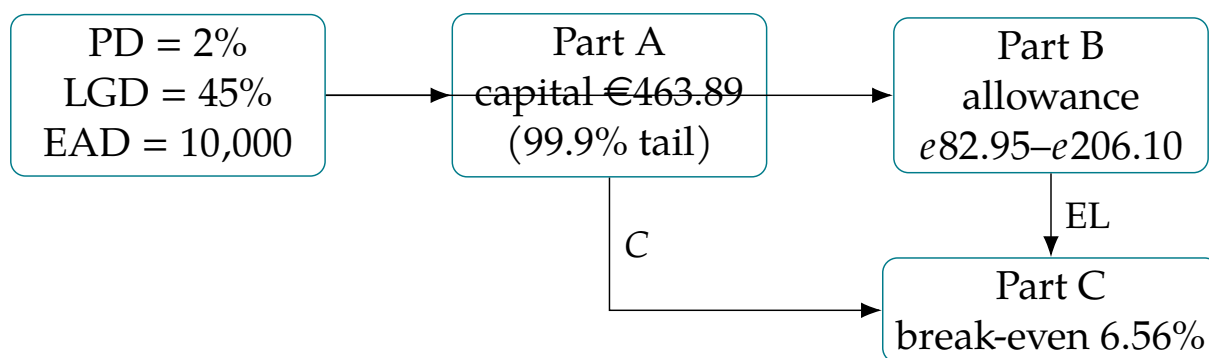


Figure 6.1: One parameter set drives capital, allowance, and price.

Chapter 7

The Companion Dashboards

The three Python scripts implement the mathematics of this book as self-contained Flask dashboards (Python 3.12; dependencies per `requirements_py312.txt`; runnable in Spyder via F5).

Script	Port	Demonstrates
<code>RWA_visualised.py</code>	5001	risk weight vs PD; K and R vs PD; RWA vs EAD; reproduces Tables 3.2 and 3.3.
<code>Impairment_visualised.py</code>	5002	stage-1 vs stage-2 allowance; scenario weighting; per-year ECL decomposition; reproduces Tables 4.1 and 4.2.
<code>Pricing_visualised.py</code>	5003	break-even rate decomposition; RAROC vs candidate rate with hurdle line; capital multiplier; EL basis switch; reproduces Table 5.2.

Recommended study sequence: reproduce Section 3.7 in `RWA_visualised.py`, then vary the PD slider and compare with Table 3.2; move to `Impairment_visualised.py` and switch the stage selector to see the cliff; finish with `Pricing_visualised.py`, first with the capital multiplier at 1.0, then at 4.0.

Chapter 8

Exercises

1. Derive the factor 12.5 from the 8% rule without looking at Chapter 3. Explain in one sentence why the factor contains no economics.
2. Compute $R(\text{PD})$ for $\text{PD} = 4\%$ by hand (two decimal places) and verify with `RWA_visualised.py`.
3. Show algebraically that the two expressions $\Phi[(G(\text{PD}) + \sqrt{R} G(0.999))/\sqrt{1-R}]$ and (3.12) are identical.
4. A supervisor raises the stress confidence level from 99.9% to 99.99%. Explain qualitatively (or compute) what happens to K for our running example.
5. Compute the stage-1 and stage-2 allowances of the running example with $m = (3\%, 3.5\%, 4\%)$ and interpret the change versus Table 4.1.
6. Prove or disprove: shifting weight from the base to the adverse scenario always increases the weighted ECL.
7. Using Table 5.2, estimate the break-even rate for capital multiplier 3.0 and $h = 14\%$ (interpolate), then verify with `Pricing_visualised.py`.
8. Explain why the RAROC of the running example at a customer rate of 8.5% exceeds 40%, and why this does not imply that retail lending earns a 40% return on equity in practice.

Hints. (1) Pure algebra: division by 0.08. (3) Split the numerator over $\sqrt{1-R}$. (4) $G(0.9999) \approx 3.719$ instead of 3.090 — the stress PD rises further. (6) True — by linearity in the adverse scenario's weight, since the adverse ECL exceeds the base ECL.

Appendix A

Notation

Symbol	Meaning
PD	probability of default (one year)
LGD	loss given default
EAD	exposure at default
R	asset correlation (systematic weight squared)
Y, ε	systematic / idiosyncratic standard normal factors
$\Phi, G = \Phi^{-1}$	standard normal CDF / quantile function
K	regulatory capital factor per unit of EAD
RW	risk weight, $12.5 \times K$
RWA	risk-weighted assets
T	remaining life (years)
m_t	marginal (conditional-on-survival) one-year PD in year t
E_t	amortised-cost balance at the start of year t
EIR	effective interest rate
ECL	expected credit loss (allowance)
EL	expected loss
B	average outstanding balance over the loan life
r_L, r_F	customer rate / funding rate
C	allocated (regulatory or economic) capital
h	hurdle rate (cost of equity)
RAROC	risk-adjusted return on capital

Appendix B

Source Notes and References

References

- [1] Regulation (EU) No 575/2013 (CRR), Art. 153 (risk-weight functions, corporates) and Art. 154 (retail exposures, correlation formula). EUR-Lex: <https://eur-lex.europa.eu/eli/reg/2013/575/oj/eng>; Art. 154 text: <https://www.legislation.gov.uk/eur/2013/575/article/154>.
- [2] Basel Committee on Banking Supervision (2005), “An Explanatory Note on the Basel II IRB Risk Weight Functions”, BIS: <https://www.bis.org/bcbs/irbriskweight.pdf>.
- [3] Gordy, M. B. (2003), “A risk-factor model foundation for ratings-based bank capital rules”, *Journal of Financial Intermediation* 12, 199–232.
- [4] Vasicek, O. (2002), “The Distribution of Loan Portfolio Value”, *Risk* 15(12), 160–162; Merton, R. C. (1974), “On the Pricing of Corporate Debt”, *Journal of Finance* 29, 449–470.
- [5] BIS Financial Stability Institute, “IFRS 9 and expected loss provisioning — Executive Summary”: <https://www.bis.org/fsi/fsisummaries/ifrs9.pdf>; IFRS 9 *Financial Instruments*, IFRS Foundation (ifrs.org), paras 5.5.15–5.5.20.
- [6] Risk-adjusted return on capital (RAROC): https://en.wikipedia.org/wiki/Risk-adjusted_return_on_capital; risk-based pricing: https://en.wikipedia.org/wiki/Risk-based_pricing.
- [7] User-provided materials: `old_chat.txt` (prior worked derivation of the RWA example) and `requirements_py312.txt` (target Python environment).

All web sources accessed October 2026. Credibility ratings follow in the Source Notes table.

Source Notes

Source	Credibility	Last updated
EUR-Lex, Regulation (EU) 575/2013	5/5	2013 (consolidated ongoing)
Legislation.gov.uk, CRR Art. 154	5/5	2013 (EU-exit amendments noted)
BIS, Explanatory Note on Basel II IRB Functions	5/5	2005
BIS FSI, IFRS 9 Executive Summary	5/5	2017
Wikipedia, RAROC	3/5	—
Wikipedia, Risk-based pricing	3/5	—

Caveats. Article numbering and the correlation constants were verified against the retrieved CRR Art. 154 text; the CRR2 output-floor timeline and CRR3 changes were not re-verified against the current consolidated text and are stated qualitatively only. The IFRS 9 paragraph references follow the IFRS Foundation numbering and the BIS FSI summary.

Open Questions

1. What is the bank's actual (supervisor-approved) PD and LGD for this borrower segment? Unobservable from outside; only the bank can answer.
2. Economic capital for a concentrated retail book exceeds the €463.89 regulatory figure — by how much requires the bank's internal tail model.
3. The current CRR3 output-floor percentage applicable to the bank's IRB portfolio changes effective RWA; not modelled here.
4. The servicer's actual stage-2 migration rules (days-past-due triggers, cure rules) materially affect IFRS 9 P&L timing.

Recommendations and Next Steps

1. Run the three companion scripts and reproduce every table in this book; each script prints its parameter set and results.
2. Stress the parameters: PD up to 20%, LGD to 90%, and observe the concavity of the risk weight and the near-linearity of the ECL.
3. Read the primary texts: CRR Arts. 153–154 and the BIS explanatory note for Part A; IFRS 9 paras 5.5.15–5.5.20 for Part B.
4. Extend the pricing script to economic capital ($3\text{--}5\times$ regulatory capital for retail) and observe the break-even rate converge toward observed market rates.

This textbook was created with AI assistance; all numerical examples were computed and cross-checked with executable code. It is an educational treatment with didactic simplifications, not regulatory, accounting, or financial advice.